The documentation and process conversion measures necessary to comply with this document shall be completed by 25 August 2004.

INCH-POUND

MIL-PRF-19500/393E
25 May 2004
SUPERSEDING
MIL-PRF-19500/393D

* PERFORMANCE SPECIFICATION SHEET

* SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON POWER,
TYPES 2N3418, 2N3418S, 2N3418U4, 2N3419, 2N3419S, 2N3419U4, 2N3420,
2N3420U4, 2N3420S, 2N3421, 2N3421S, AND 2N3421U4,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1. Scope. This specification covers the performance requirements for NPN, silicon, transistors for use in medium power switching applications. Four levels of product assurance are provided for each device type, and two levels of product assurance for die (element evaluation) are provided, as specified in MIL-PRF-19500.

* 1.2. Physical dimensions. See figure 1 (similar to TO-5 for long leaded devices and TO-39 for short leaded devices), figure 2 and figure 3 for JANHC and JANKC (die), and figure 4 (2N3418U4 through 2N3421U4) dimensions.

* 1.3. Maximum ratings.

Type	P_T T_A = +25°C (1)	P_T T_C = +100°C (1)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_C (2)	T_{STG} and T_J	$R_{\theta JA}$	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W	°C/W
2N3418, 2N3418S	1.0	5	85	60	8	3	5	-65 to	175	18
2N3418U4	N/A	15	85	60	8	3	5	+200	N/A	4.5
2N3419, 2N3419S	1.0	5	125	80	8	3	5	-65 to	175	18
2N3419U4	N/A	15	125	80	8	3	5	+200	N/A	4.5
2N3420, 2N3420S	1.0	5	85	60	8	3	5	-65 to	175	18
2N3420U4	N/A	15	85	60	8	3	5	+200	N/A	4.5
2N3421, 2N3421S	1.0	5	125	80	8	3	5	-65 to	175	18
2N3421U4	N/A	15	125	80	8	3	5	+200	N/A	4.5

* (1) For derating, see figure 5 through figure 7.
* (2) This value applies for $t_p \leq 1$ ms, duty cycle ≤ 50 percent.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://www.dodssp.daps.mil/.

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* 1.4. Primary electrical characteristics at T_A = +25°C.

Limits	h_{FE2} (1)		h_{FE4} (1)		$V_{CE(sat)}$ (1)	$V_{BE(sat)}$ (1)	$\|h_{fe}\|$	C_{obo}
	V_{CE} = 2 V dc I_C = 1 A dc		V_{CE} = 5 V dc I_C = 5 A dc		I_C = 1 A dc I_B = 0.1 A dc	I_C = 1 A dc I_B = 0.1 A dc	V_{CE} = 10 V dc I_C = 0.1 A dc f = 20 MHz	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz
	2N3418 2N3418S 2N3418U4 2N3419 2N3419S 2N3419U4	2N3420 2N3420S 2N3418U4 2N3421 2N3421S 2N3421U4	2N3418 2N3418S 2N3418U4 2N3419 2N3419S 2N3419U4	2N3420 2N3420S 2N3418U4 2N3421 2N3421S 2N3421U4				
					V dc	V dc		pF
Min	20	40	10	15		0.6	1.3	
Max	60	120			0.25	1.2	8	150

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

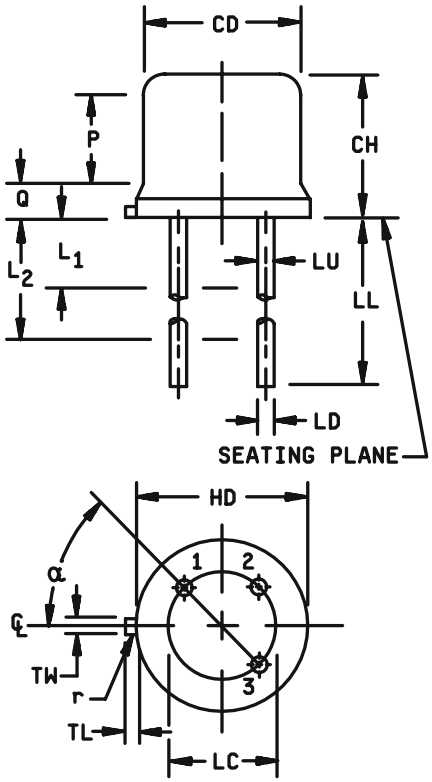
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://www.dodssp.daps.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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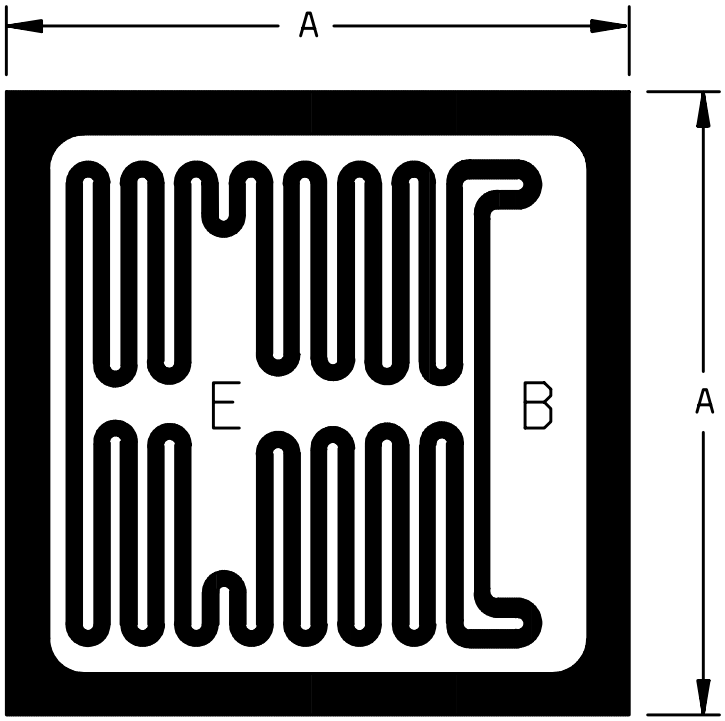
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	
LL	.500	.750	12.7	19.05	7
LU	See notes 7, 13, 14				
L1		.050		1.27	7
L2	.250		6.35		7
P	.100		2.54		5
Q		.040		0.86	4
TL	.029	.045	0.74	1.14	3,10
TW	.028	.034	0.71	.86	9,10
r		.010		0.25	11
α	45° TP		45° TP		6



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. Lead number three is electrically connected to case.
9. Beyond r maximum, TW shall be held for a minimum length of .021 inch (0.53 mm).
10. Lead number 4 omitted on this variation.
11. Symbol r applied to both inside corners of tab.
12. For transistor types 2N3418S, 2N3419S, 2N3420S, 2N3421S, LL is .500 (12.70 mm) minimum and .750 (19.05 mm) maximum.
13. For transistor types 2N3418, 2N3419, 2N3420, 2N3421, LL is .500 (38.10 mm) minimum, and 1.750 (44.45 mm) maximum.
14. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
15. Lead 1 is emitter, lead 2 is base, and lead 3 is collector.

* FIGURE 1. Physical dimensions.

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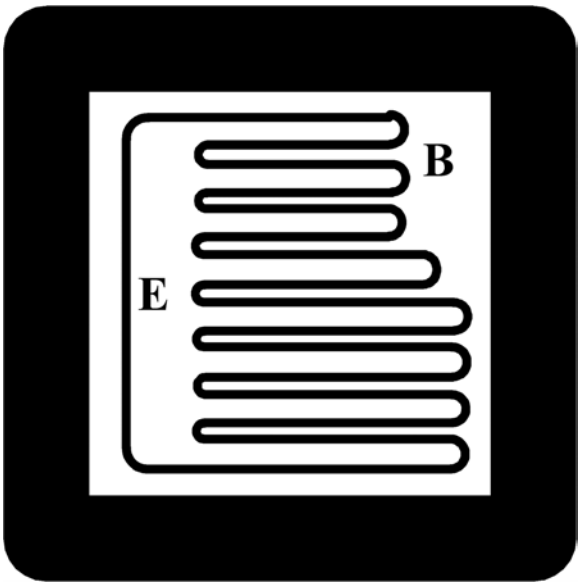


Letter	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.117	.127	2.97	3.23

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is .005 (0.13 mm).
4. The physical characteristics of the die are;
   Thickness: .008 (0.20 mm) to .012 (0.30 mm), tolerance is .005 (0.13 mm).
   Top metal: Aluminum, 40,000 Å minimum, 50,000 Å nominal.
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = .015 (0.38 mm) x .0072 (.183).
   E = .015 (0.38 mm) x .0060 (.152).

FIGURE 2. JANHCA and JANKCA die dimensions.

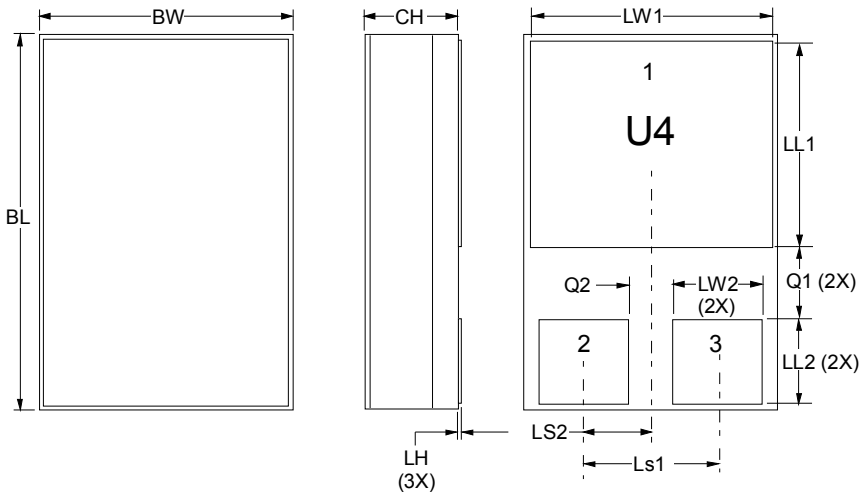
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1. Chip size: .075 x .075 inch ±.002 inches (1.905 x 1.905 mm ±0.051 mm).
2. Chip thickness: .010 ±.0015 inches nominal (0.254 ±0.0381 mm).
3. Top metal: Aluminum 30,000 Å minimum, 33,000 Å nominal.
4. Back metal: A. Al/Ti/Ni/Ag12kÅ/3kÅ/7kÅ/7kÅmin.,15kÅ/5kÅ/10kÅ/10kÅ nom.
   B. Gold 2,500Å minimum, 3000Å nominal.
5. Backside: Collector.
6. Bonding pad: B = .023 x .008 inch (0.5842 x 0.2032 mm), E = .049 x .008 inch (1.2446 x 0.2032 mm).

FIGURE 3. JANHC and JANKC B-version die dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.72
BW	.145	.155	3.68	3.94
CH	.050	.070	1.27	1.77
LH	.010	.020	0.26	0.50
LW1	.135	.145	3.43	3.68
LW2	.047	.057	1.19	1.45
LL1	.085	.125	2.16	3.17
LL2	.045	.075	1.14	1.91
LS1	.065	.095	1.65	2.41
LS2	.033	.048	.825	1.21
Q1	.005	.085	0.12	2.16
Q2	.008	.048	.200	1.22
TERM 1	Collector			
TERM 2	Base			
TERM 3	Emitter			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.

* FIGURE 4. Physical dimensions and configuration (U4).

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, (similar to TO-5 and TO-39), figures 2 and figure 3 (die), and figure 4 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

* 4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and, tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein shall be performed by the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
7	Optional	Optional
9	I_{CEX1} and h_{FE2}	I_{CEX1}
11	I_{CEX1}; h_{FE2}; ΔI_{CEX1} = 100 percent or 50 nA dc, whichever is greater; Δh_{FE2} = +15 , -10 percent change of initial value.	I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100 percent or 100 nA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100 percent or 50 nA dc, whichever is greater; Δh_{FE2} = +15, -10 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent or 100 nA dc, whichever is greater; Δh_{FE2} = +20, -10 percent of initial value.
14	Required	Required

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +175°C minimum and a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3.

4.3.2 Screening JANHC or JANKC. Screening of die shall be in accordance with MIL-PRF-19500.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The $Z_{\theta JX}$ limit used in screen 3c shall comply with the thermal impedance graph in figure 8 through figure 10 (less than or equal to the curve value at the same t_H time) and/or shall be less than the process determined statistical maximum limit as outlined in method 3131.

a. I_M measurement current 10 mA.

b. I_H forward heating current 2 A (min).

c. t_H heating time 10 ms.

d. t_{md} measurement delay time 50 μs max.

e. V_{CE} collector-emitter voltage 10 V dc minimum.

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 55°C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) shall be in accordance with group A, subgroup 2. Delta requirements shall be in accordance with table III and the notes for table III herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Test condition A. All internal wires for each device shall be pulled separately.
B4	1037	V_{CE} = 5 V dc, 2,000 cycles.
B5	1027	V_{CE} = 5 V dc, P_T adjusted to achieve T_J and time required in MIL-PRF-19500.
B7	3053	T_A = +25°C, I_B = 0.5 A dc, I_C = 3.0 A dc, see figure 11.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and /or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 - 30 V dc, power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0.
2	1039	HTRB (high temperature reverse bias): Test condition A, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANJ, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table III and the notes for table III herein.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6	1026	T_A = +25 ±5°C; T_J = +150°C minimum. (Not applicable to JAN, JANTX,and JANTXV). V_{CB} = 40 V dc for types 2N3418, 2N3418S, 2N3420, and 2N3420S. V_{CB} = 60 V dc for types 2N3419, 2N3419S, 2N3421, and 2N3421S.

4.4.3.2 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 1.0 A dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. The measuring current magnitude shall be 1 mA dc.

d. Reference temperature measuring point shall be the case.

e. Reference point temperature shall be +25°C ≤ T_R ≤ +75°C and recorded before the test is started.

f. Mounting arrangement shall be out to heat sink.

g. Maximum limits for $R_{\theta JC}$ shall be 6.67°C/W.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Thermal impedance	3101	See 4.3.3	$Z_{\theta JX}$		55	°C/W
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 50 mA dc, I_B = 0, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3418, 2N3418S 2N3420, 2N3420S 2N3418U4, 2N3420U4				60		V dc
2N3419, 2N3419S 2N3421, 2N3421S 2N3419U4, 2N3421U4				80		V dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -0.5 V dc	I_{CEX1}			
2N3418, 2N3418S 2N3420, 2N3420S 2N3418U4, 2N3420U4		V_{CE} = 80 V dc			0.3	µA dc
2N3419, 2N3419S 2N3421, 2N3421S 2N3419U4, 2N3421U4		V_{CE} = 120 V dc			0.3	µA dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter cutoff current	3041	Bias condition D; I_B = 0	I_{CEO}			
2N3418, 2N3418S 2N3420, 2N3420S 2N3418U4, 2N3420U4		V_{CE} = 45 V dc			5.0	µA dc
2N3419, 2N3419S 2N3421, 2N3421S 2N3418U4, 2N3421U4		V_{CE} = 60 V dc			5.0	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc, I_C = 0	I_{EBO1}		0.5	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 8 V dc, I_C = 0	I_{EBO2}		10	µA dc
Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 100 mA dc, pulsed (see 4.5.1)	h_{FE1}			
2N3418, 2N3418S 2N3418U4, 2N3419, 2N3419S, 2N3419U4				20		
2N3420, 2N3420S, 2N3420U4, 2N3421, 2N3421S, 2N3421U4				40		
Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1.0 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N3418, 2N3418S, 2N3418U4, 2N3419, 2N3419S, 2N3419U4				20	60	
2N3420, 2N3420S, 2N3420U4, 2N3421, 2N3421S, 2N3421U4				40	120	
Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 2 A dc, pulsed (see 4.5.1)	h_{FE3}			
2N3418, 2N3418S, 2N3418U4, 2N3419, 2N3419S, 2N3419U4				15		
2N3420, 2N3420S, 2N3420U4, 2N3421, 2N3421S, 2N3421U4				30		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward current transfer ratio 2N3418, 2N3418S, 2N3418U4, 2N3419, 2N3419S, 2N3419U4 2N3420, 2N3420S, 2N3420U4, 2N3421, 2N3421S, 2N3421U4	3076	V_{CE} = 5 V dc; I_C = 5 A dc, pulsed (see 4.5.1)	h_{FE4}	10 15		
Base-emitter voltage (saturated)	3066	Test condition A; I_C = 1.0 A dc, I_B = 0.1 A dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$ 6/	0.6	1.2	V dc
Base-emitter voltage (saturated)	3066	Test condition A; I_C = 2.0 A dc, I_B = 0.2 A dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$ 6/	0.7	1.4	V dc
Saturation voltage and resistance (collector-emitter)	3071	I_C = 1.0 A dc, I_B = 0.1 A dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$ 6/		0.25	V dc
Saturation voltage and resistance (collector-emitter)	3071	I_C = 2.0 A dc, I_B = 0.2 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$ 6/		0.5	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -0.5 V dc	I_{CEX2}			
2N3418, 2N3418S 2N3420, 2N3420S		V_{CE} = 80 V dc			50	µA dc
2N3419, 2N3419S 2N3421, 2N3421S		V_{CE} = 120 V dc			50	µA dc
Low-temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1 A dc pulsed (see 4.5.1)	h_{FE5}	10		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio magnitude of common emitter)	3306	V_{CE} = 10 V dc; I_C = 0.1 mA dc; f = 20 MHz	$\|h_{fe}\|$	1.3	8	

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Open-circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0 , 100 kHz ≤ f ≤ 1 MHz	C_{obo}		150	pF
Switching time		I_C = 1.0 A dc, $I_{B(1)}$ = 100 mA dc, $I_{B(2)}$ = -100 mA dc,	t_r t_d t_s		0.22 0.08 1.10	µs µs µs
* Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +100°C, t ≥ 1 s, 1 cycle, see figure 13				
Test 1		I_C = 3 A dc, V_{CE} = 5 V dc				
Test 2		I_C = 0.4 A dc, V_{CE} = 37 V dc				
Test 3						
2N3418, 2N3418S 2N3418U4, 2N3420, 2N3420S, 2N3420U4		I_C = 0.185 A dc, V_{CE} = 60 V dc				
2N3419, 2N3419S 2N3419U4, 2N3421, 2N3421S, 2N3421U4		I_C = 0.12 A dc, V_{CE} = 80 V dc				
Safe operating area (clamped switching)	3053	T_A = +25°C, I_B = 0.5 A dc, I_C = 3.0 A dc, see figure 14				
Electrical measurements		See subgroup 2 herein				
Subgroup 6 and 7						
Not applicable						

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Measured at a point on the leads no further than .125 inch (3.18 mm) from the case.

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* TABLE II. Group E inspection (all quality levels) - for qualification and re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc.	
Electrical measurements		See table I, subgroup 2.	
Subgroups 3			3 devices c = 0
Destructive physical analysis (DPA)	2102		
Subgroup 4			15 devices, c = 0
Thermal resistance	3131	The following applies for qualification for $R_{\theta JA}$ and $R_{\theta JC}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (can apply to multiple specification sheets).	
Thermal impedance, thermal resistance curves		Each supplier shall submit their (typical) design thermal impedance curves. In addition, test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			3 devices c = 0
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V, Condition B for devices < 400 V.	

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TABLE III. Groups B and C delta measurements. 1/ 2/ 3/

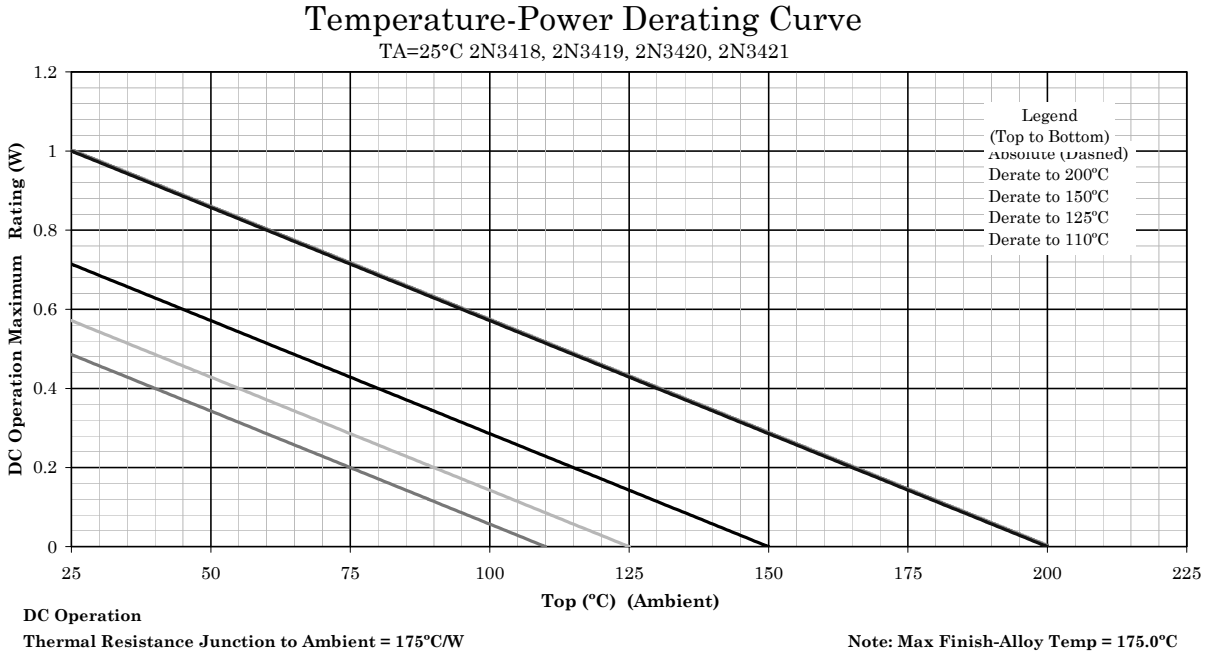
Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward-current transfer ratio	3076	V_{CE} = 2 V dc I_C = 1.0 A dc pulsed (see 4.5.1)	Δh_{FE2} 1/		+20 percent, -10 percent change from initial group A reading	
2.	Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -0.5 V dc	ΔI_{CEX1} 1/			
	2N3418, 2N3418S 2N3420, 2N3420S		V_{CE} = 80 V dc			100 percent of initial value or 100 nA dc, whichever is greater	
	2N3419, 2N3419S 2N3421, 2N3421S		V_{CE} = 120 V dc			100 percent of initial value or 100 nA dc, whichever is greater	
3.	Saturation voltage and resistance (collector-emitter)	3071	I_C = 1.0 A dc, I_B = 0.1 A dc, pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$ 1/		±50 mV dc change from previously measured value	

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
a. Subgroup 4, see table III herein, steps 1 and 3.
b. Subgroup 5, see table III herein, steps 1, 2, and 3.

2/ The delta measurements for 4.4.2.2 (JAN, JANTX, JANTXV) of MIL-PRF-19500 are as follows: Steps 1, 2, and 3 of table III herein, after each step in 4.4.2.2.

3/ The delta measurements for table VII (JANS) of MIL-PRF-19500 are as follows: Subgroups 2, 3, and 6, see table III herein, steps 1, 2, and 3.

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NOTES:

1/ Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.

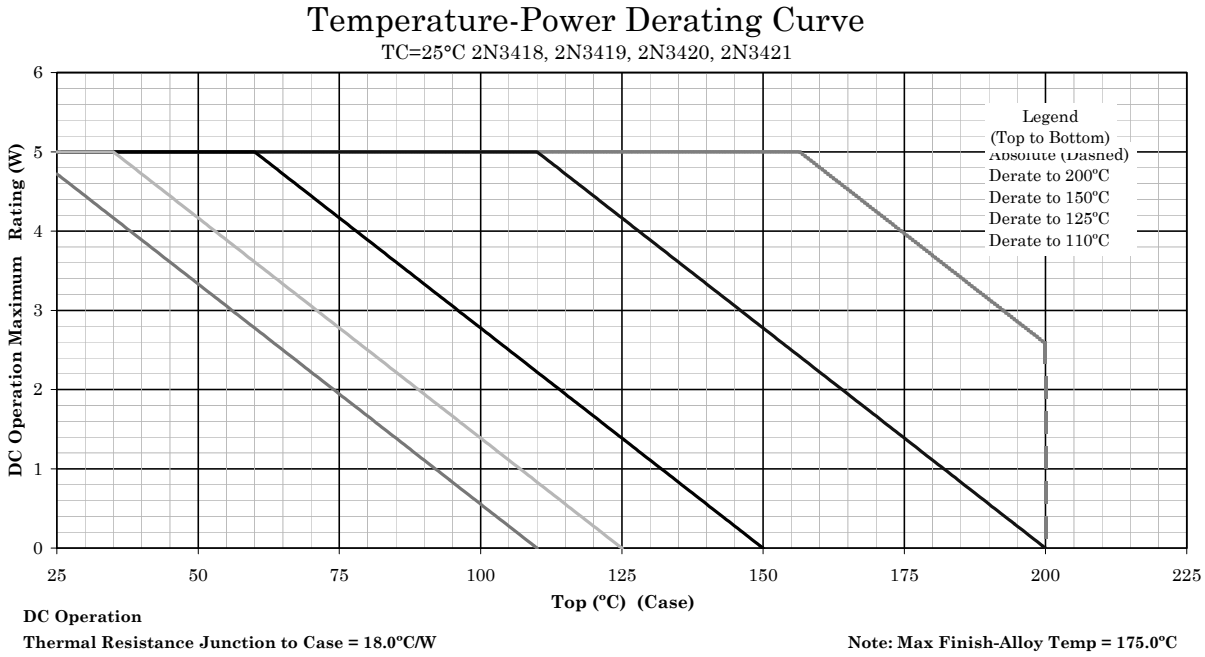
2/ Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3)

3/ Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.

4/ Derate design curve chosen at $T_J \leq +125°C$ and +110°C, to show power rating where most users want to limit T_J in their application.

* FIGURE 5. Derating for 2N3418, 2N3418S, 2N3419, 2N3419S, 2N3420, 2N3420S, 2N3421, and 2N3421S ($R_{\theta JA}$) leads .375 inch PCB (TO-5 and TO-39).

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NOTES:

1/ Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.

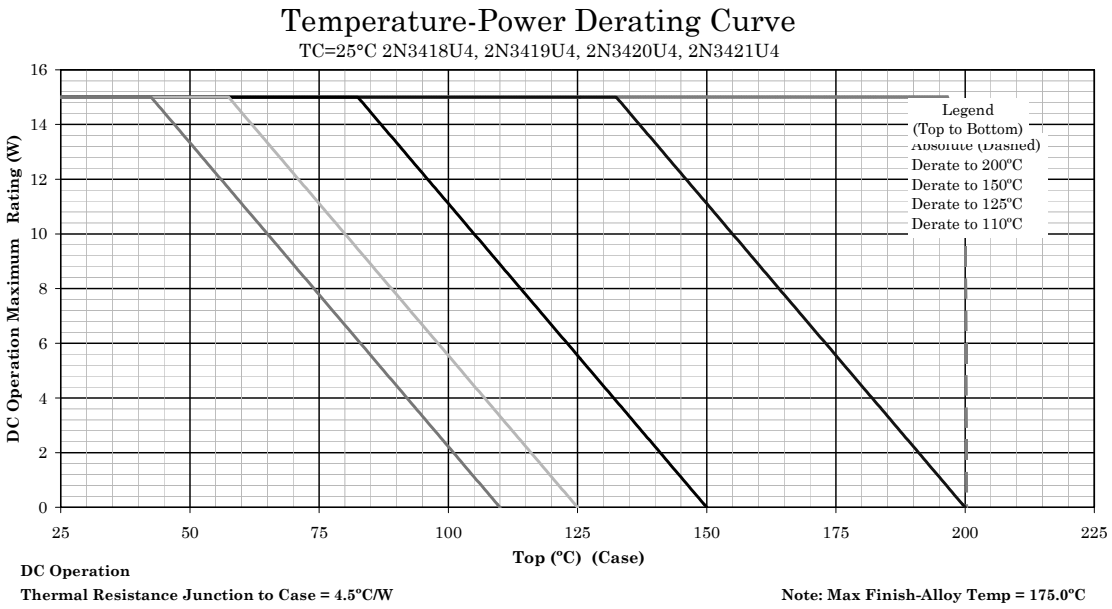
2/ Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3)

3/ Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.

4/ Derate design curve chosen at T_J ≤ +125°C and +110°C, to show power rating where most users want to limit T_J in their application.

* FIGURE 6. Derating for 2N3418, 2N3418S, 2N3419, 2N3419S, 2N3420, 2N3420S, 2N3421, and 2N3421S ($R_{\theta JC}$) (TO-5 and TO-39).

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NOTES:

1/ Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.

2/ Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3)

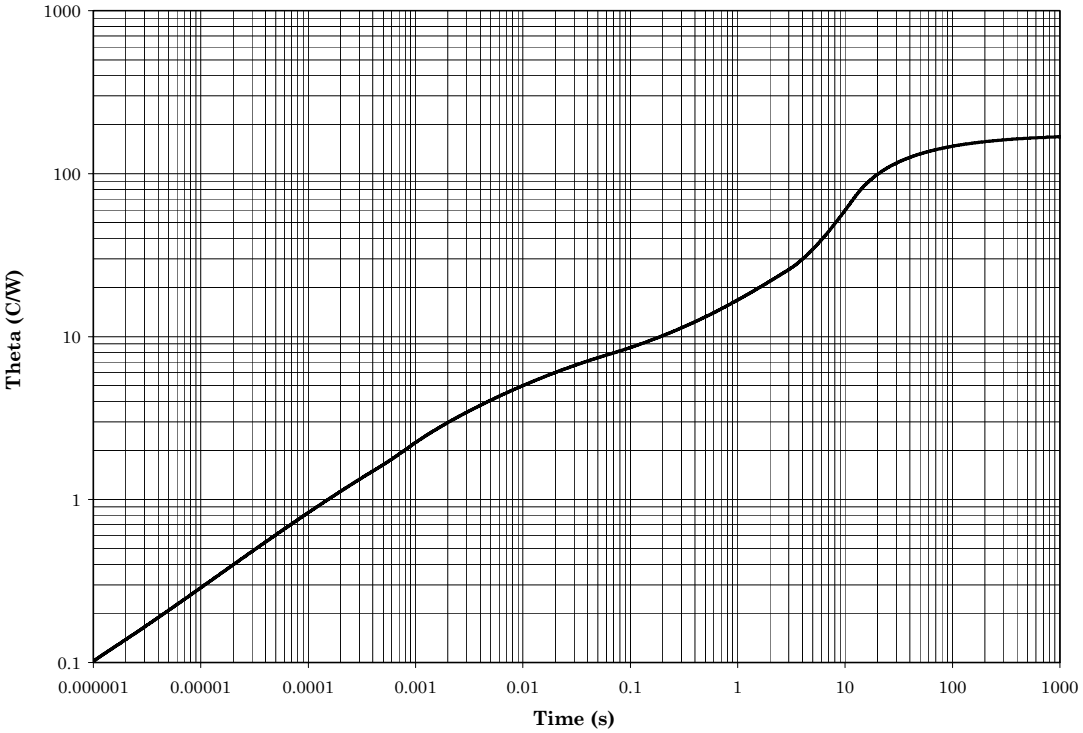
3/ Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.

4/ Derate design curve chosen at T_J ≤ +125°C and +110°C, to show power rating where most users want to limit T_J in their application.

* FIGURE 7. Derating for 2N3418U4, 2N3419U4, 2N3420U4, and 2N3421U4 ($R_{\theta JC}$) (U4).

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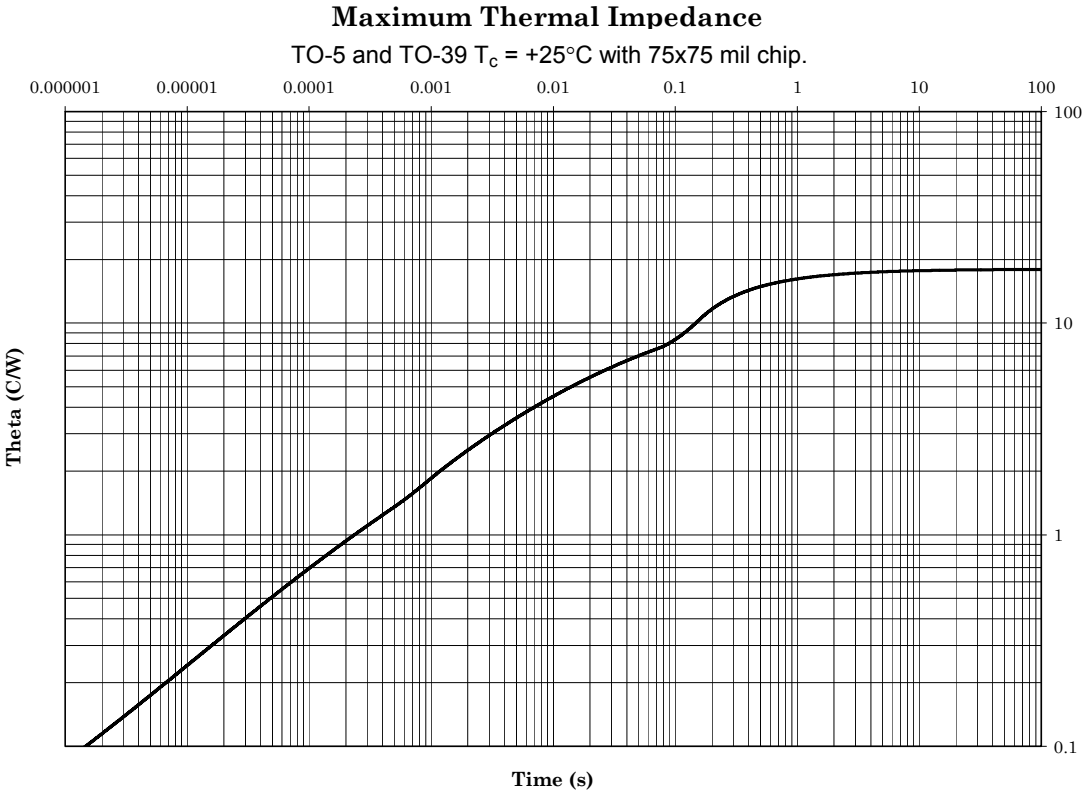
Maximum Thermal Impedance



T_A = +25°C, P_{diss} = 800mW, Thermal Resistance $R_{\theta JA}$ = 175°C/W

* FIGURE 8. Thermal impedance graph ($R_{\theta JA}$) for 2N3418, 2N3418S, 2N3419, 2N3419S, 2N3420, 2N3420S, 2N3421, and 2N3421S leads .375 inch PCB (TO-5 and TO-39).

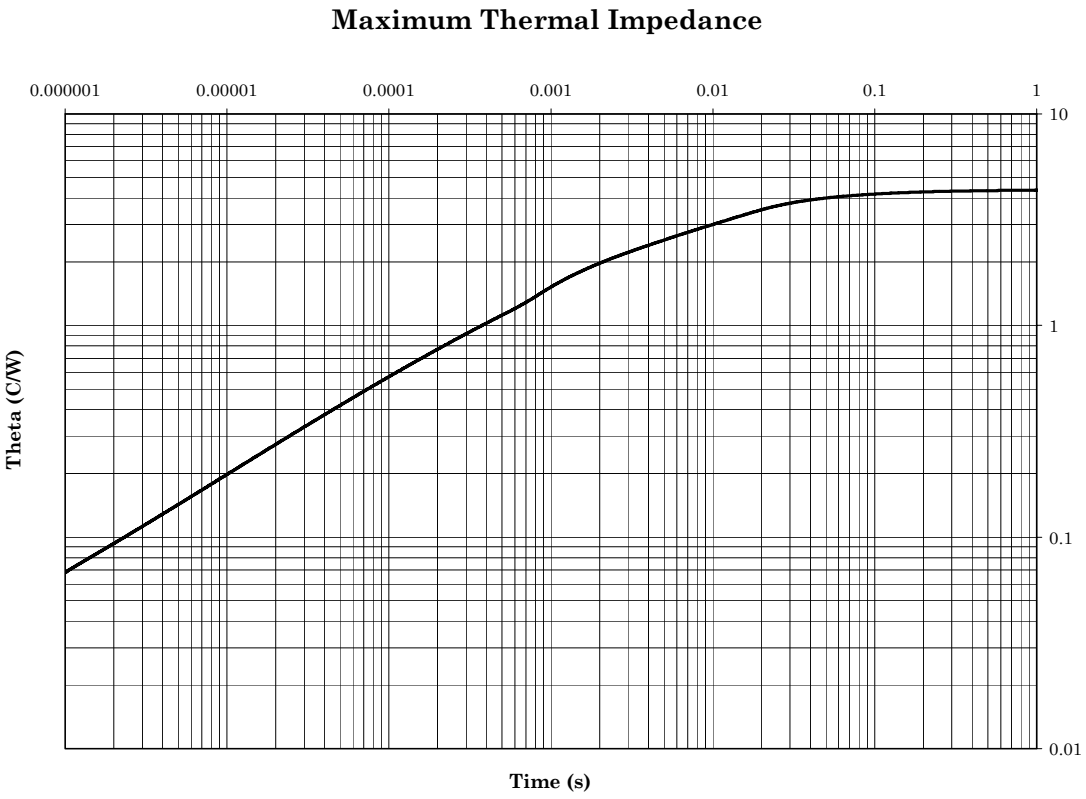
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T_c = +25°C, Thermal Resistance $R_{\theta JC}$ = 18°C/W

FIGURE 9. Thermal impedance graph ($R_{\theta JC}$) for 2N3418, 2N3418S, 2N3419, 2N3419S, 2N3420, 2N3420S, 2N3421, and 2N3421S leads .375 inch PCB (TO-5 and TO-39).

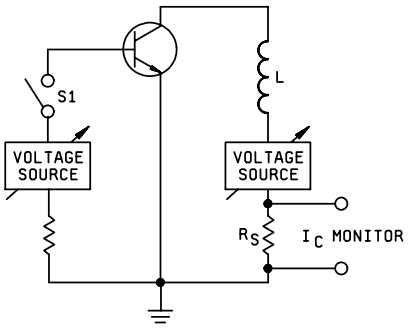
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T_c = +25°C, Thermal Resistance $R_{\theta JC}$ = 18°C/W

* FIGURE 10. Thermal impedance graph ($R_{\theta JC}$) for 2N3418U4, 2N3419U4, 2N3420U4, and 2N3421U4 PCB (U4).

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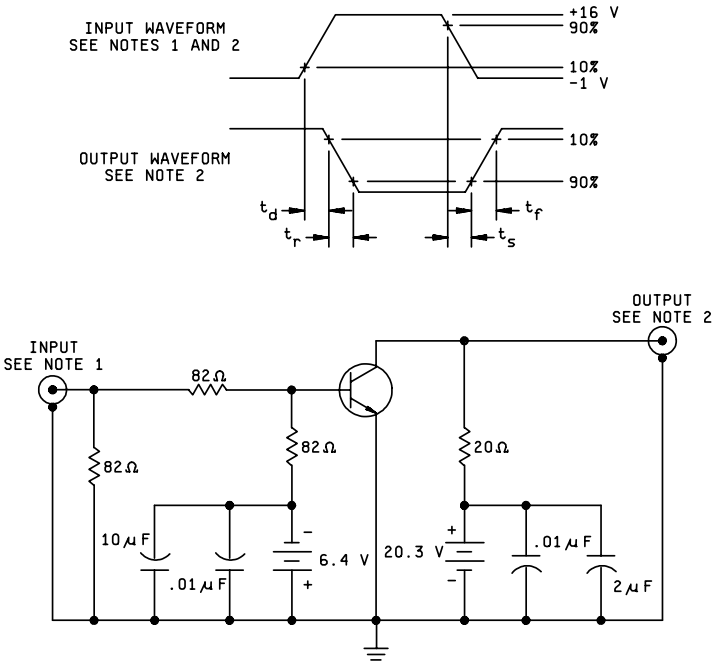
NOTES:

$R_S \leq 1.0\ \Omega$ (noninductive), L = 10 mH

Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1.
3. Perform specified end-point tests.

FIGURE 11. Unclamped inductive sweep test circuit diagram.



NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 2 μs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored by an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $R_{in} \geq 10\ M\Omega$, $C_{in} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The DC power supplies may require additional by-passing in order to minimize ringing.

FIGURE 12. Pulse response test circuit.

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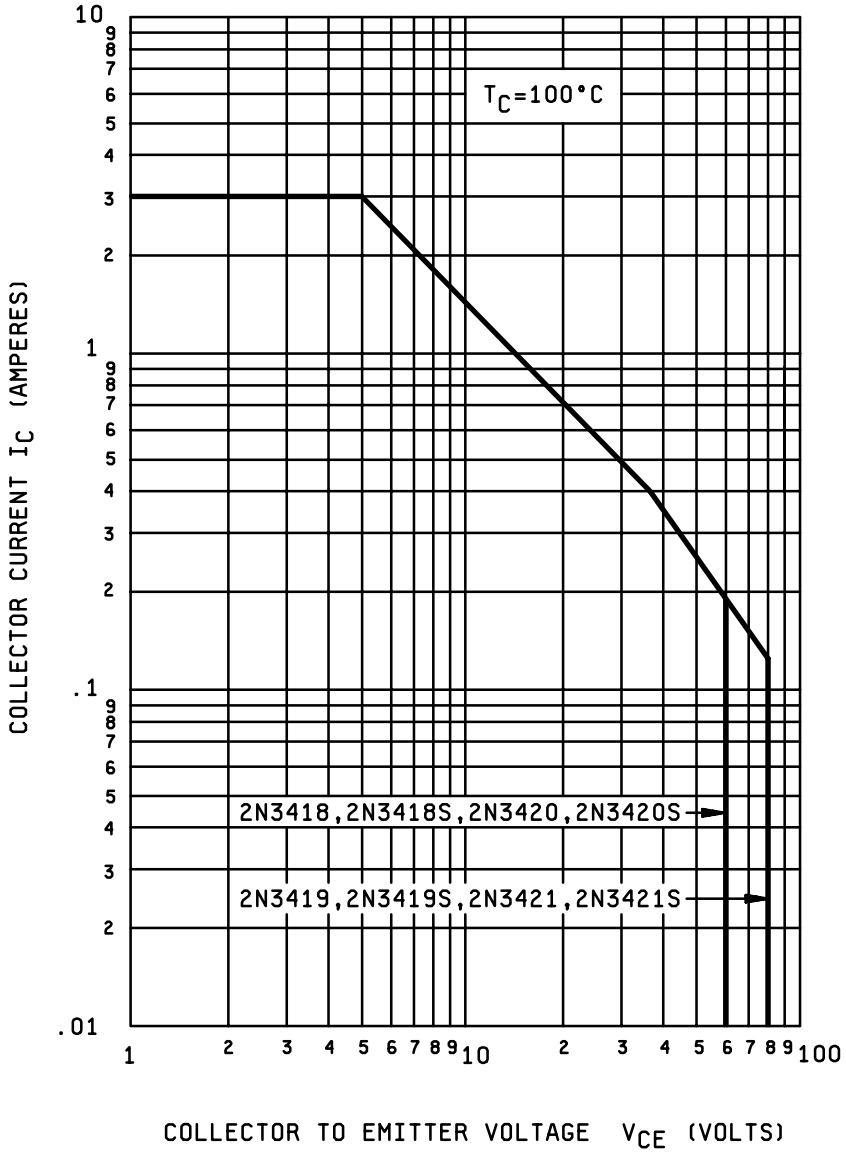
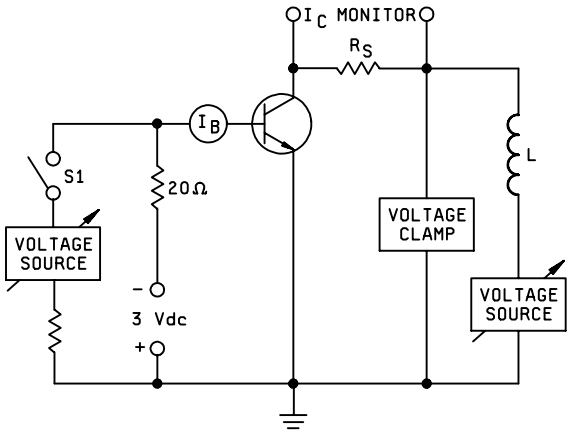


FIGURE 13. Maximum safe operating region.

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NOTES:
Voltage clamp: 2N3418, 2N2418S, 2N3420, 2N3420S = 85 V dc
2N3419, 2N3419S, 2N3421, 2N3421S = 125 V dc

$R_S \leq 1.0\ \Omega$ (noninductive), L = 40 mH

Procedure:
1. With switch S_1 closed, set the specified test conditions.
2. Open S_1.
3. Perform specified end-point tests.

* FIGURE 14. Clamped inductive sweep test circuit diagram.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML No. 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000 or e-mail vqe.chief@dla.mil.

6.4. Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N5152) will be identified on the QML.

JANHC and JANKC ordering information		
PIN	Manufacturer	
	33178	43611
2N3418 2N3419 2N3420 2N3421	JANHCA2N3418 JANHCA2N3419 JANKCA2N3420 JANKCA2N3421	JANHCB2N3418 JANHCB2N3419 JANKCB2N3420 JANKCB2N3421
2N3418S 2N3419S 2N3420S 2N3421S	JANKCA2N3418S JANKCA2N3419S JANKCA2N3420S JANKCA2N3421S	JANKCB2N3418S JANKCB2N3419S JANKCB2N3420S JANKCB2N3421S

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* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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